Agile Software Development with SCRUM

www.scrumguides.com

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Today's Agenda

Opening: program overview, knowing each other

Understanding Agile and SCRUM

SCRUM simulation

Extra SCRUM topics

Concepts of Agile Planning

We will be having 10-minute breaks each 60-90 minutes.

About myself

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SCRUM guides

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www.scrumguides.com

Our presence

- Ukrainian Agile community
  www.agileukraine.org

  Join our Google discussion group

- Ukrainian SCRUM portal
  www.scrum.com.ua

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Exercise “Continuum”

Project successes/failures

- Have you been on a successful project?
- Have you been on an unsuccessful project?

Agile might help you, let's see…

- Understanding Agile and SCRUM

The two approaches, two cultures

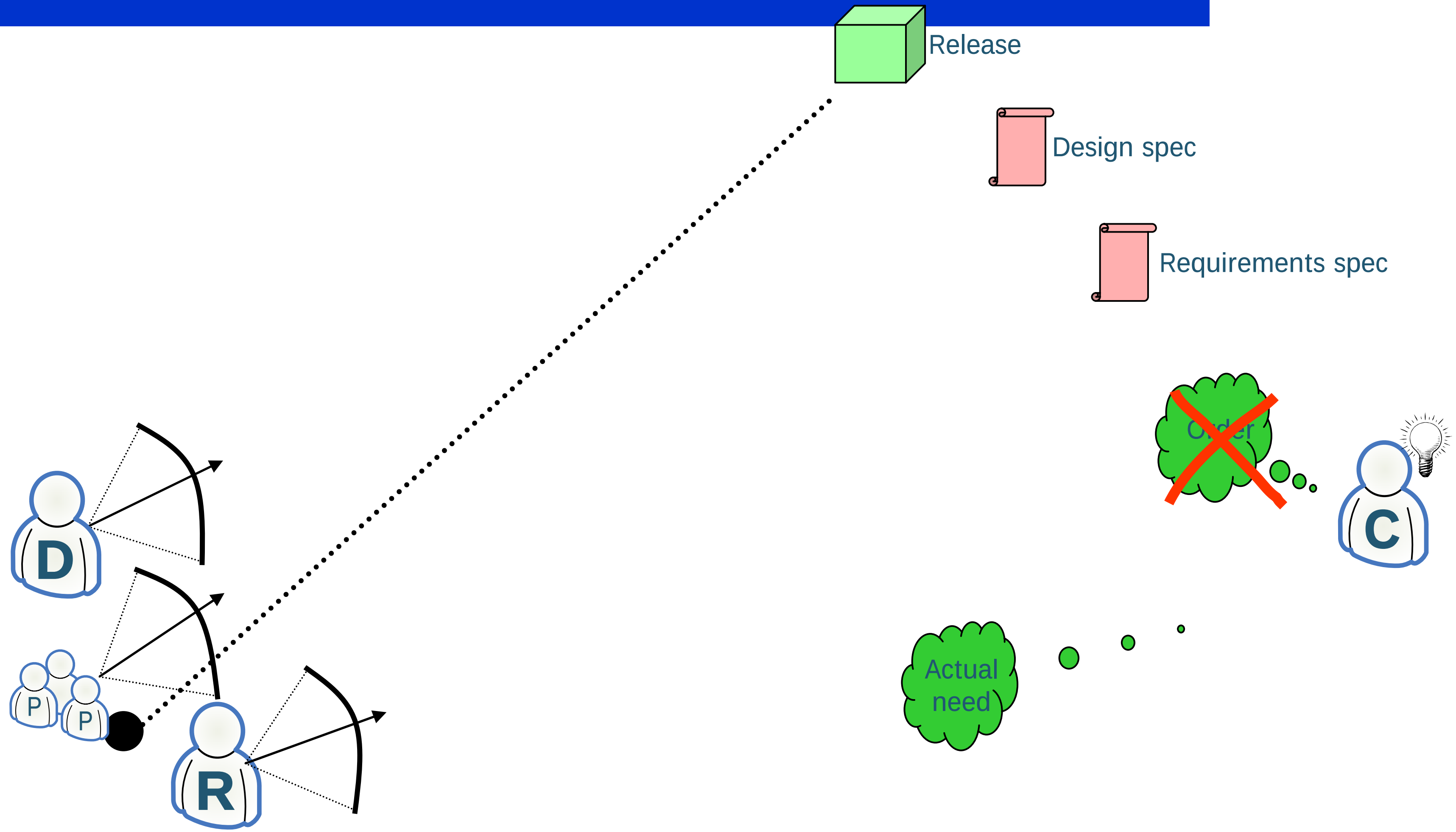
Predictive approach

Sees projects as predictive processes which results can be foreseen (predicted) with acceptable probability after reasonable planning and study efforts spent.

Adaptive approach

Questions predictability of projects due to their built-in complexity. It puts stress on steering based on observations.

The predictive approach



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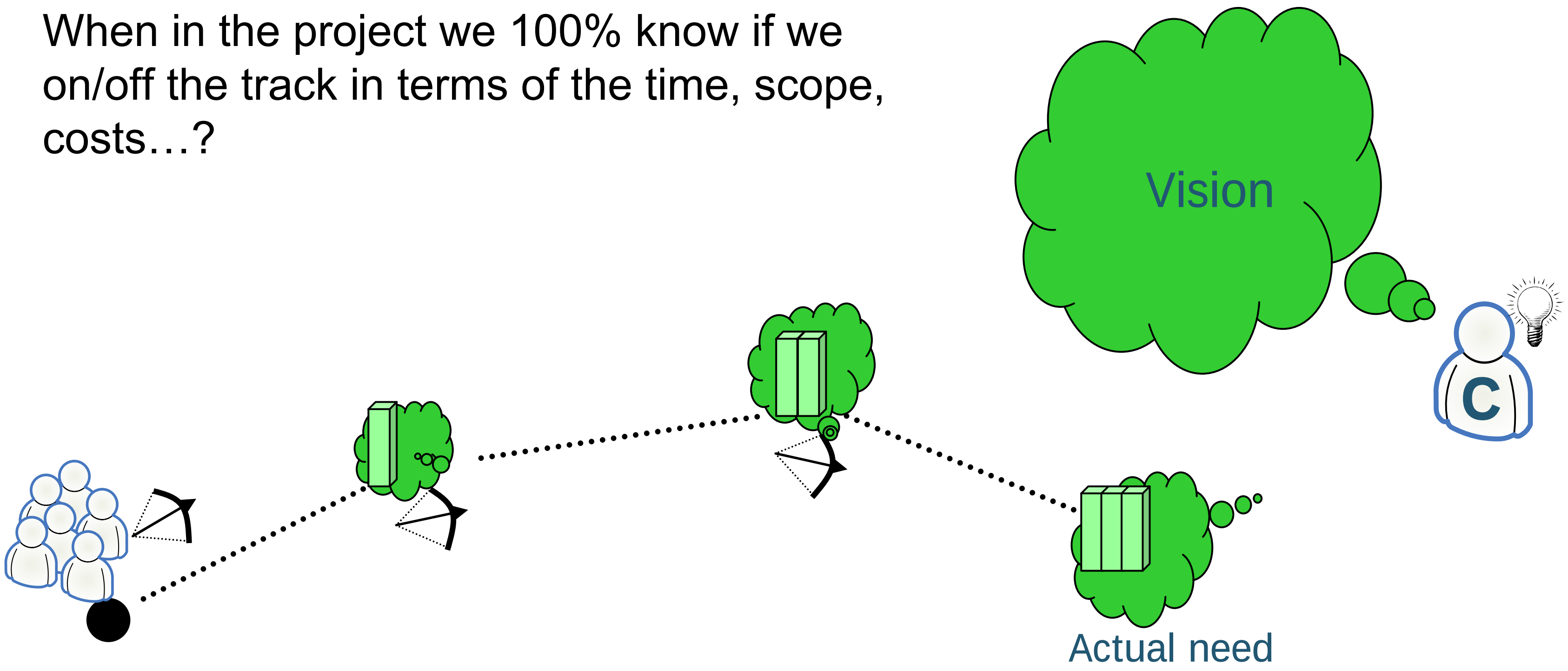
The adaptive approach

When in the project we 100% know if we on/off the track in terms of the time, scope, costs…?

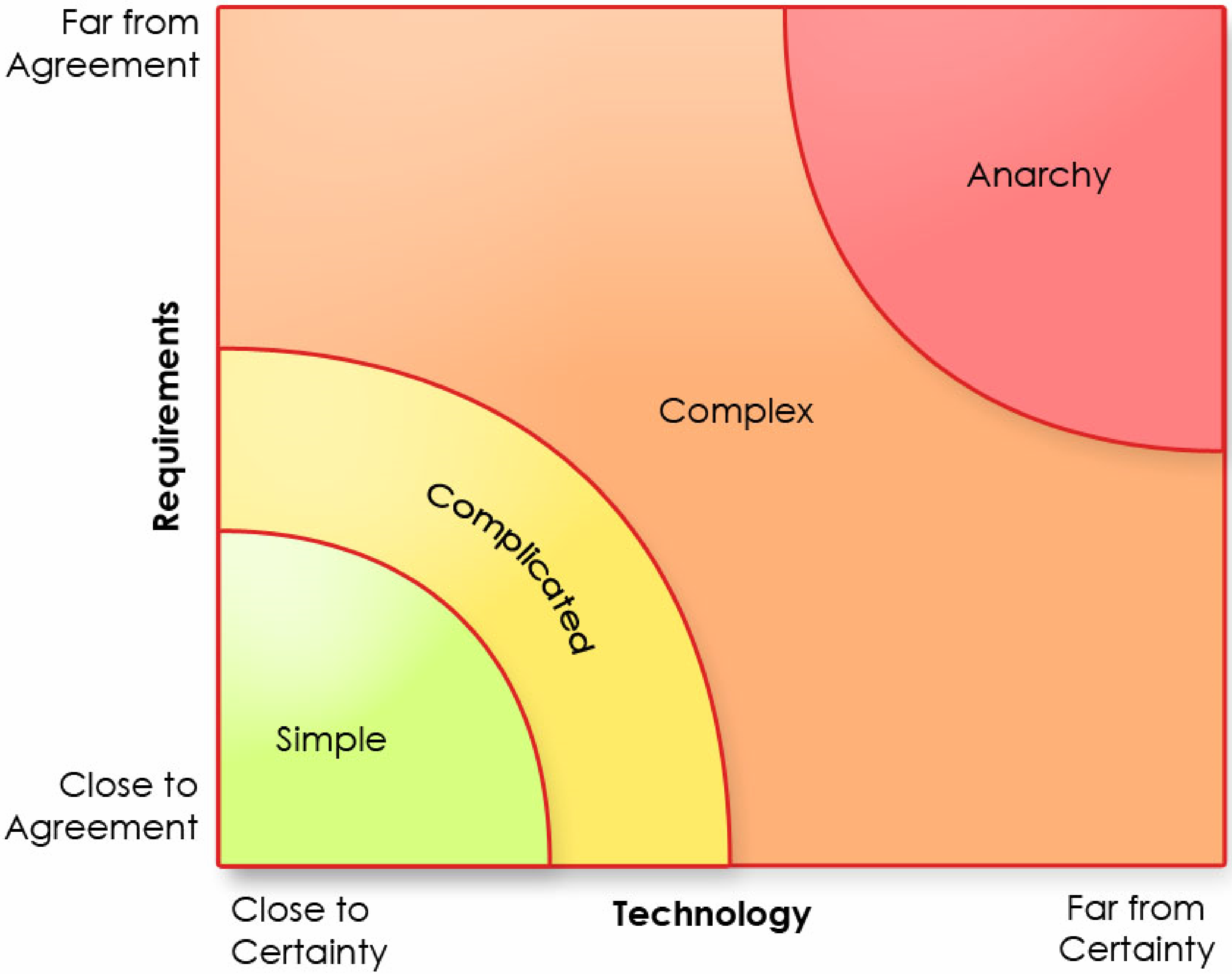


The two approaches to project management

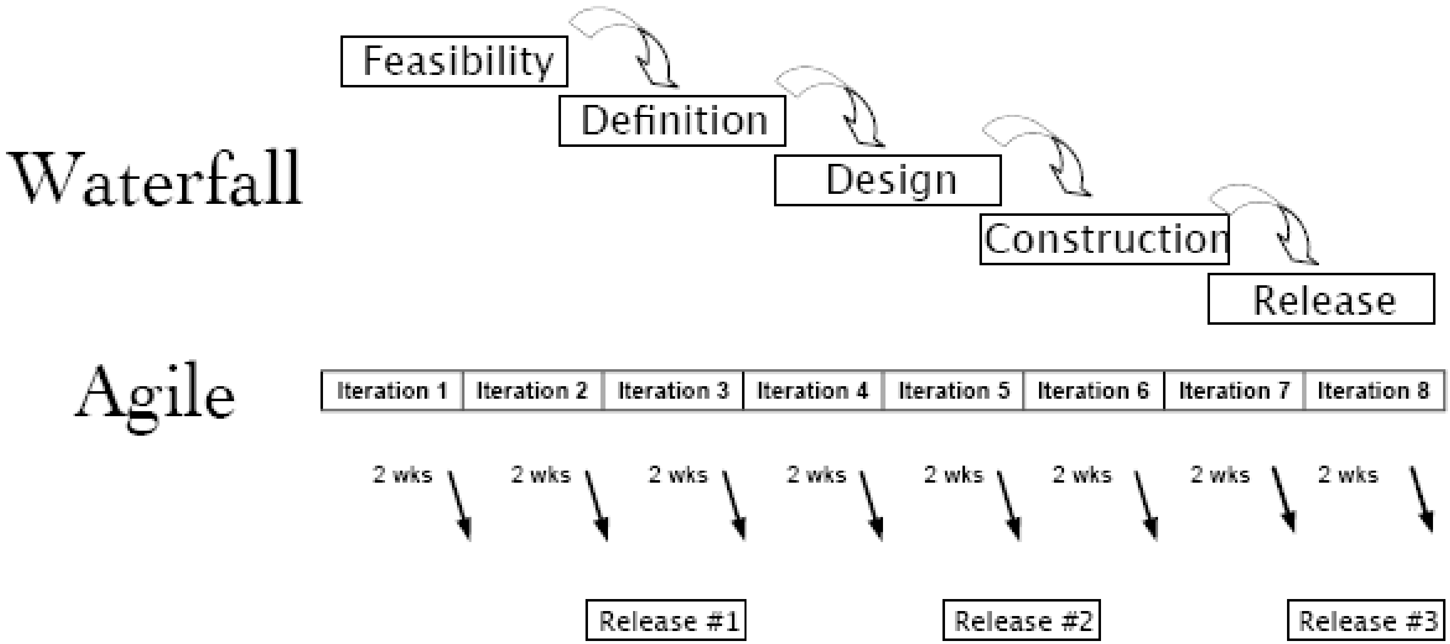
- Predictive approach
  - Heavy-weight;
  - Process-oriented;
  - Plan-driven;
  - “Waterfall”.

- Adaptive approach
  - Light-weight;
  - People-oriented;
  - Value-driven;
  - “Agile”.

The Spectrum of Process Complexity



The two approaches on the time scale



Which advantages/disadvantages of the approaches you see?

Agile and SCRUM

- SCRUM – one of the Agile approaches.

  It is a project framework, or a set of recommendations by following which you can increase the chance of successful ending of your projects.

Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Main aspects of Agile

The main aspects of Agile

- ☐ **Team Work**
- ☐ Prioritization
- ☐ Short cycles
- ☐ Learn and Adapt

The main aspects of Agile: **Team work**

- What is a team?
- How to make good team?

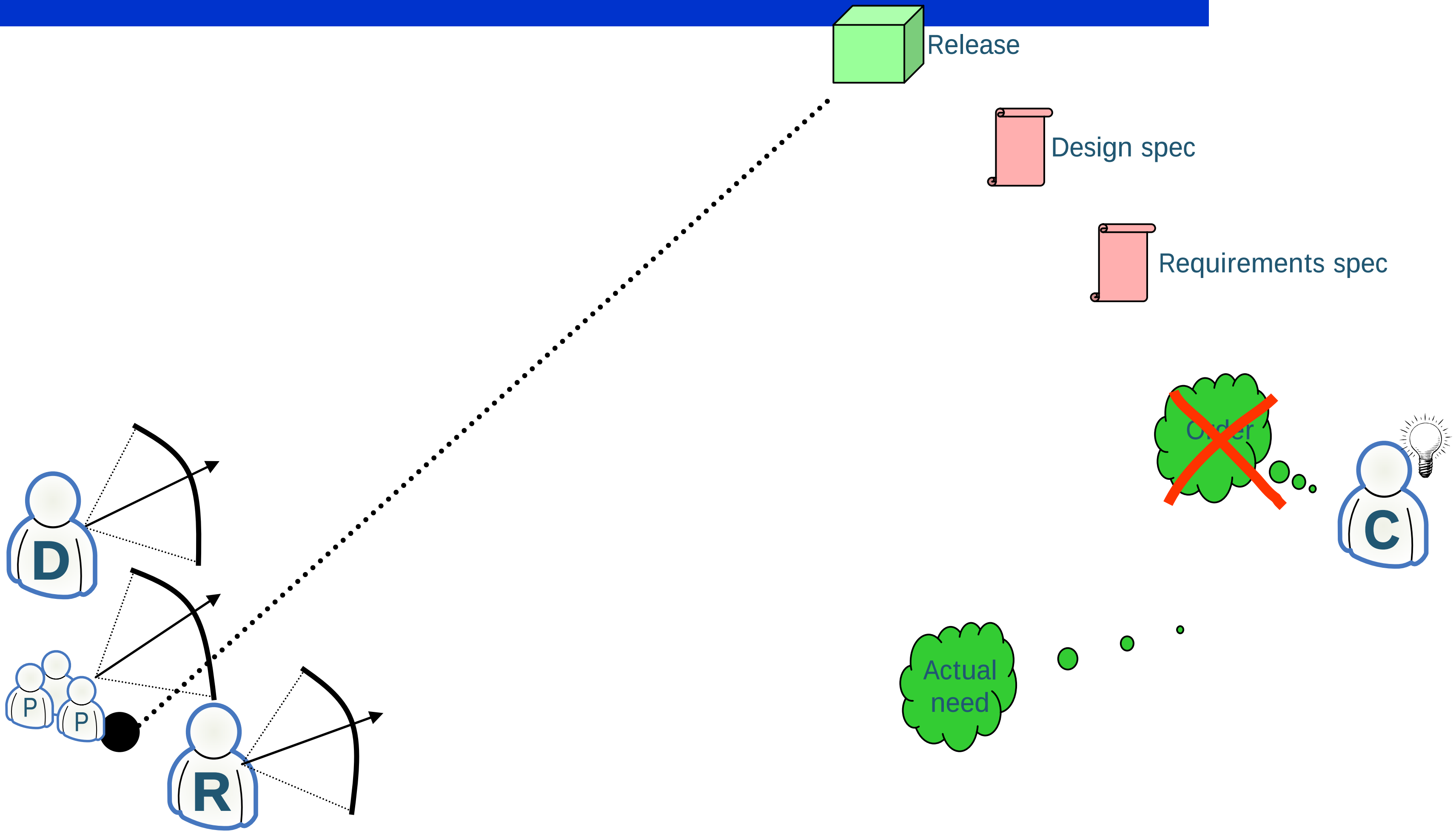
Team work (cont.)

Exercise “an airplane factory”.

Team work (cont.)

- What is a team?
- Are you in a good team?
- How to become a greater team?
- How to evaluate and motivate people's work in a team?

Is here such a thing as “team work”?



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A SCRUM team is

- A *cross-functional* group of people (5-9 members) responsible for managing itself to develop the product.

The main aspects of Agile

- ☑ Team Work
- ☐ **Prioritization**
- ☐ Short cycles
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Prioritization and Traditional approaches

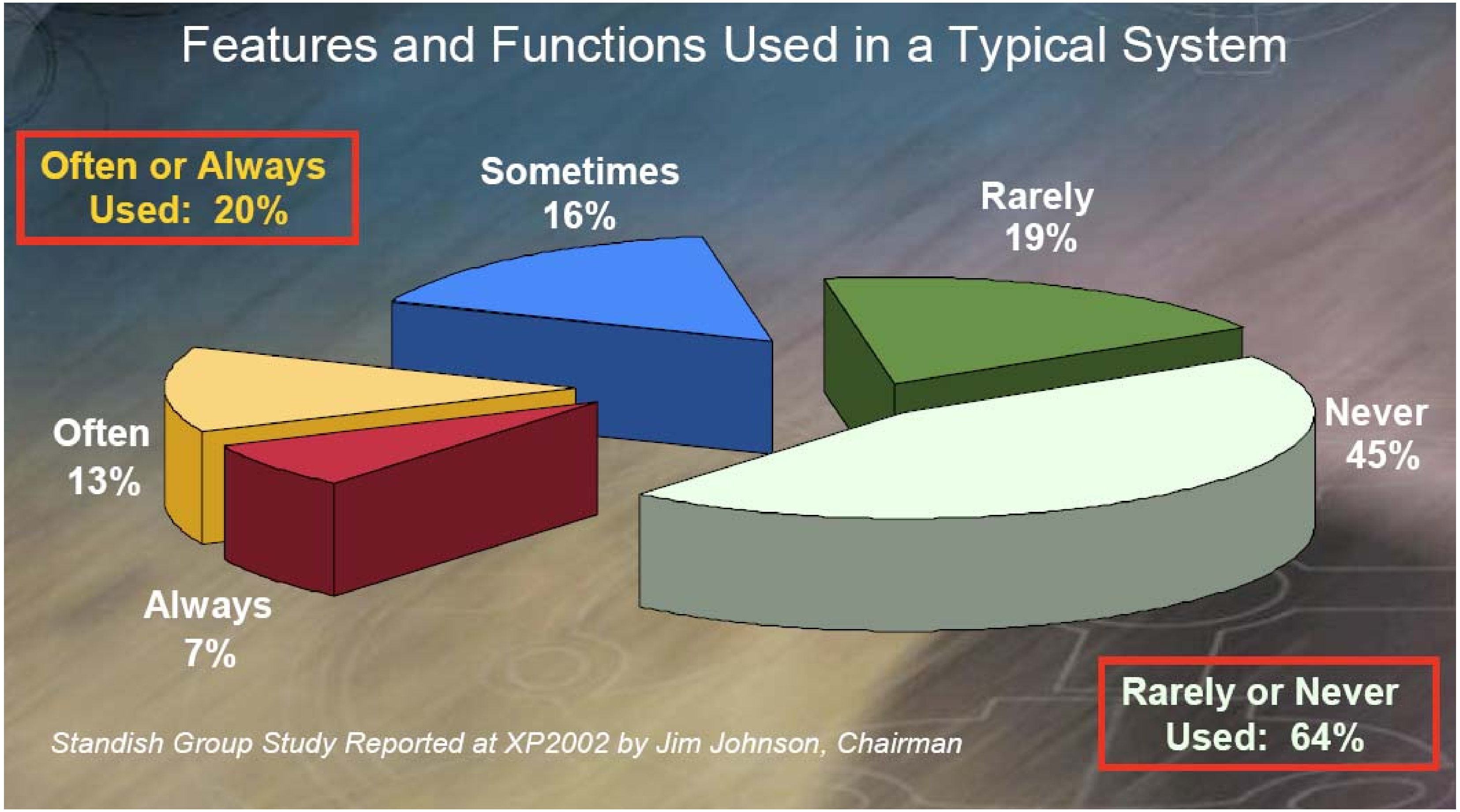
- An alternative tool to backlogs are requirement specifications.

“Gimme all requirements, or ….”

As a result ...

- As a result all requirements are the "top priority".
- Which is basically the same as not having the priorities at all.

And as a result …



What does it mean to us?

- As a consequence: we can cut costs and duration of our projects in 2/3!

- Does it mean we (the teams) will earn less money?

What about your projects?

- Think about your current (recent) projects.
- List 5-10 features that could have been avoided or simplified by keeping the product as good as it was.
- What would be your % of rarely/never used features?

Prioritize (cont.)

YAGNI - “You ain’t gonna need it”.

A strategy of postponing decisions until the last possible moment.

Product backlog

Just-in-time requirements management tool.

The “menu” (Agile) approach

- Product backlog is the project’s menu.
- “Servants” help to make better decisions.

What saves our projects?

"The features that we manage to de-scope are the savers of our projects" © Kent Beck

The main aspects of Agile

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The main aspects of Agile: **Short Cycles**

The shorter the cycle the more efficient the process

1. We have more time to “play” with the product that we are developing;
2. The sooner we can catch a defect the easier it is to fix it;
3. The shorter the cycle (the smaller a batch of work) the less need we have in creating intermediate artifacts.

Short Cycles (cont.)

“Fail fast” (c) Ken Schwaber

Iterative and Incremental approach

- How do you eat an elephant?
- One bite at a time!

Shorter Cycles = Faster Feedback

1. Daily meetings
2. Code Reviews
3. Release to end-users
4. Iteration reviews
5. Continuous integration
6. Feedback from (onsite) clients
7. Unit-testing
8. (Automated) acceptance testing
9. Pair programming

Shorter Cycles = Faster Feedback (ordered)

- Pair programming *(immediate feedback)*
- Unit-testing *(5-10 minutes)*
- Continuous integration *(hourly)*
- Feedback from (onsite) clients *(daily)*
- Daily meetings *(daily)*
- Code Reviews *(some days)*
- (Automated) acceptance testing *(some days)*
- Iteration reviews *(each second week)*
- Release to end-users *(some weeks-months)*

The main aspects of Agile

☑ Team Work

☑ Prioritization

☑ Short cycles

☐ **Learn and Adapt**

Exercise "Ball Points"

The main aspects of Agile: **Learn and Adapt**

- As we work we learn lots of new things about the product being developed, technologies being used, our clients, ourselves.
- By incorporating this information into daily work we can become better day by day.

Visibility tools

In order to evolve the way we work we need to keep everything visible to ourselves and our clients

Tools that might help:

- Sprint Burndown chart
- Task boards
- Wikis

The main aspects of Agile

- ☑ Team Work
- ☑ Prioritization
- ☑ Short cycles
- ☑ Learn and Adapt

Would SCRUM help in your case

- Think about your current (recent) project.
- List 3-5 main things that are (were) preventing it from being more successful.
- Can SCRUM help to <u>avoid</u> them?
  Can SCRUM <u>make</u> them <u>visible</u> earlier?
  How can SCRUM help you <u>solve</u> them?

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- ☐ Extra SCRUM topics
- ☐ Concepts of Agile Planning

SCRUM in 59 minutes

Game structure

Planning

- Choose a product and your PO
- Help your PO to build up a PB
- Help your PO to prioritize the PB items
- Plan your first sprint

Sprinting

- Day one - *7 min*
- Daily standup - *3 min*
- Day two - *7 min*
- Demonstration - 3 *min (per a team)*

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SCRUM recap

- Roles
  - The Team
  - ScrumMaster
  - Product Owner

- Artifacts
  - Product Backlog
  - Sprint Backlog
  - Sprint/Release Burndown Chart

- Meetings (ceremonies)
  - Sprint Planning
  - Daily Scrum ("Stand-Up")
  - Sprint Review (Demo)
  - Retrospective

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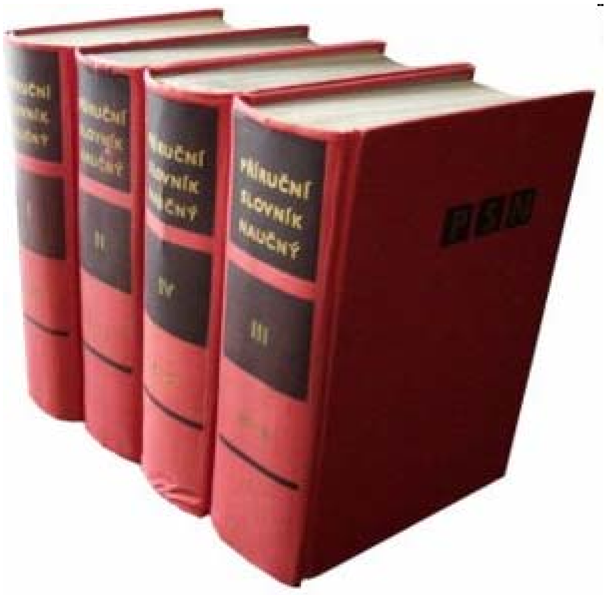
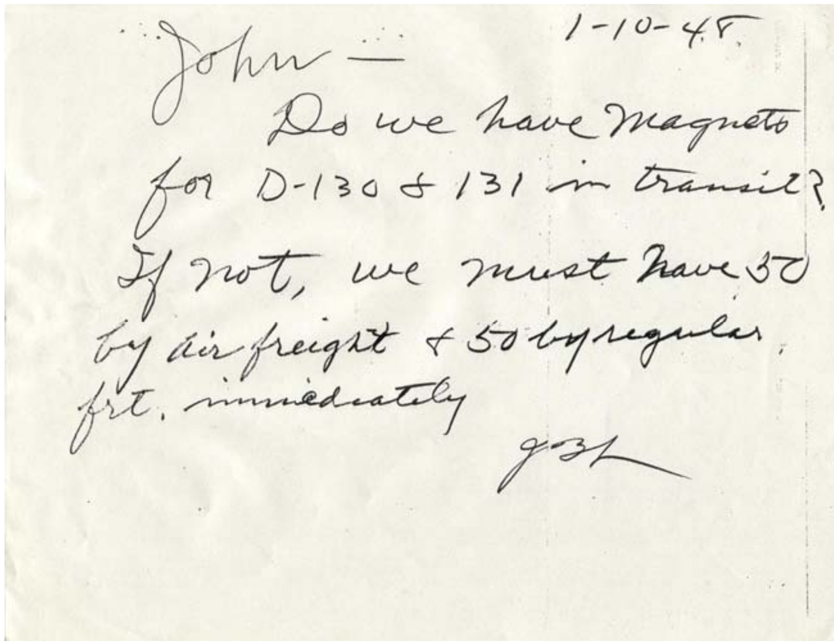
Agile Planning

- ☐ **Requirements in SCRUM**
- ☐ Levels of Planning
- ☐ Project Steering
- ☐ Estimation Techniques

Requirements

- What are the requirements to requirements?





John — 1-10-48.
Do we have magneto
for D-130 & 131 in transit?
If not, we must have 50
by air freight & 50 by regular
frt. immediately
JGL

What is important is …

- That communications/negotiation happens between the customer and the team.

- User stories help deferring the details till later
- They talk problems not solutions
- They fit nicely as your Product Backlog items

User Story is…

User stories are simple, clear, brief descriptions of functionality that will be valuable to either a user or purchaser of a product

Expressed in a form similar to:

As a <user> I can <do> so that <value>

Samples – Travel reservation system

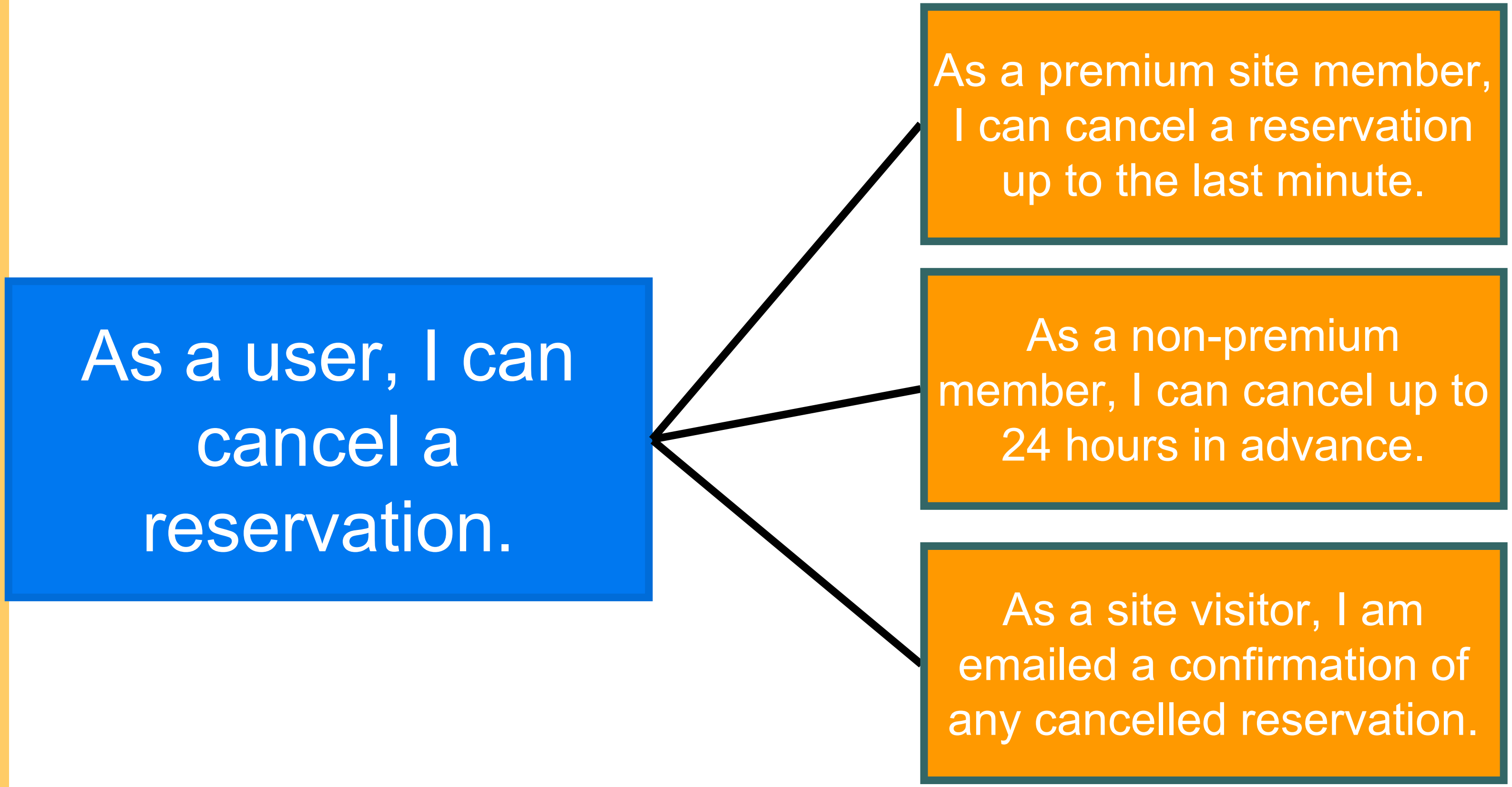
As a user, I can reserve a hotel room.

As a vacation planner, I can see photos of the hotels.

As a user, I can cancel a reservation.

As a user, I can restrict searches so that I only see hotels with available rooms.

Details added in smaller substories



Details added as tests

- High level tests are added to the story
  - Can be used to express additional details and expectations

As a user, I can cancel a reservation.

- Verify that a premium member can cancel the same day without a fee.
- Verify that a non-premium member is charged 10% for a same-day cancellation.
- Verify that an email confirmation is sent.
- Verify that the hotel is notified of any cancellation.
- Figure out what to do if the user's card is expired.

From Mike Cohn's "Agile Estimating and Planning"

Agile Planning

- ☑ Requirements in SCRUM
- ☐ **Levels of Planning**
- ☐ Project Steering
- ☐ Estimation Techniques

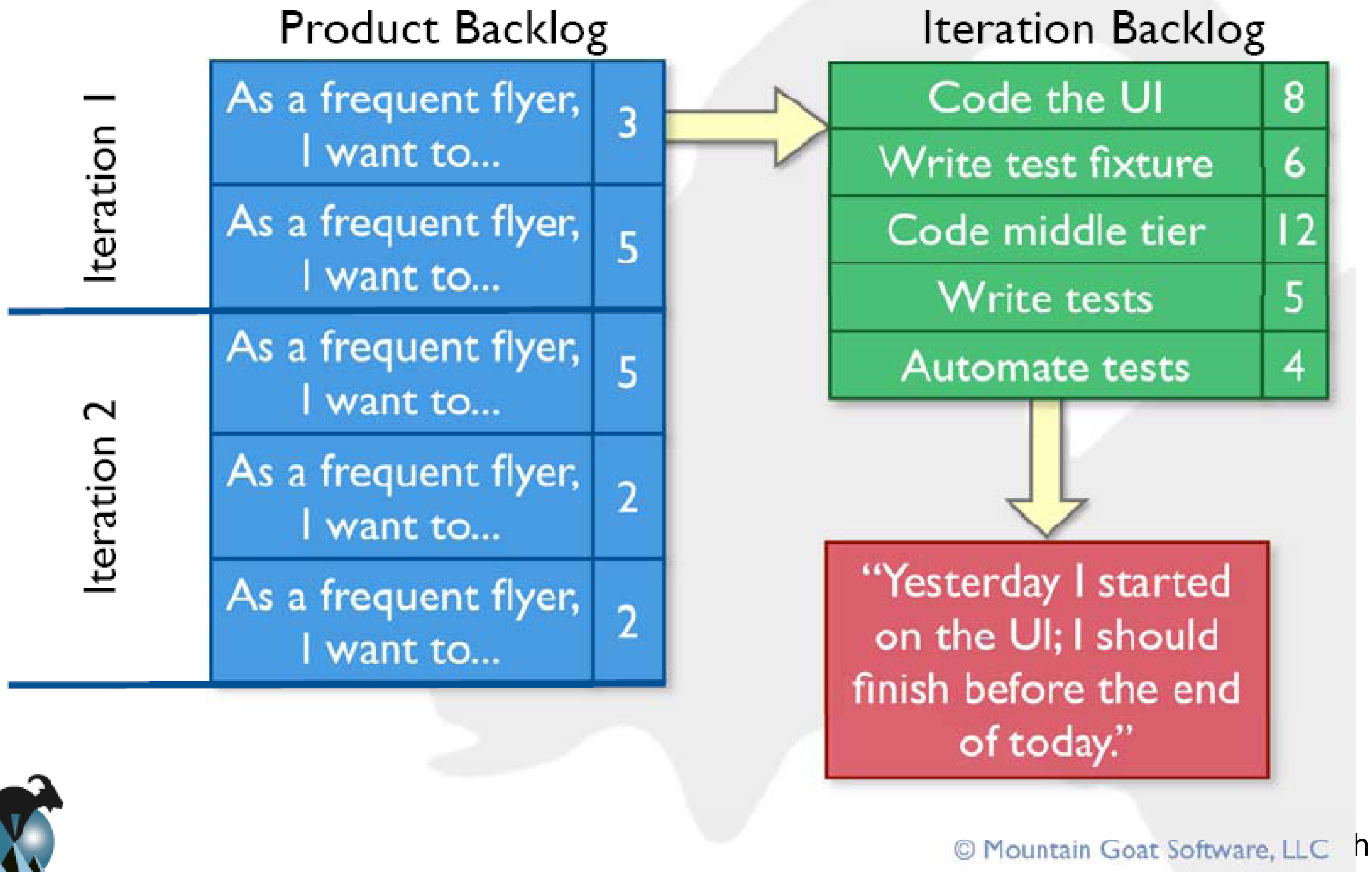
The two levels of planning

- Strategic level / Story level / product backlog
  - It is all about value. Hence the customers are at better position to control this level.
  - It is the menu of a restaurant.

- Tactical level / Task level / spring backlog
  - The tasks are defined in technical jargon. So the teams control control this level.
  - It is the kitchen of a restaurant. Do you really want to look inside? :)

From Mike Cohn's "Planning Agile Projects"

Relating the different planning levels



Agile Planning

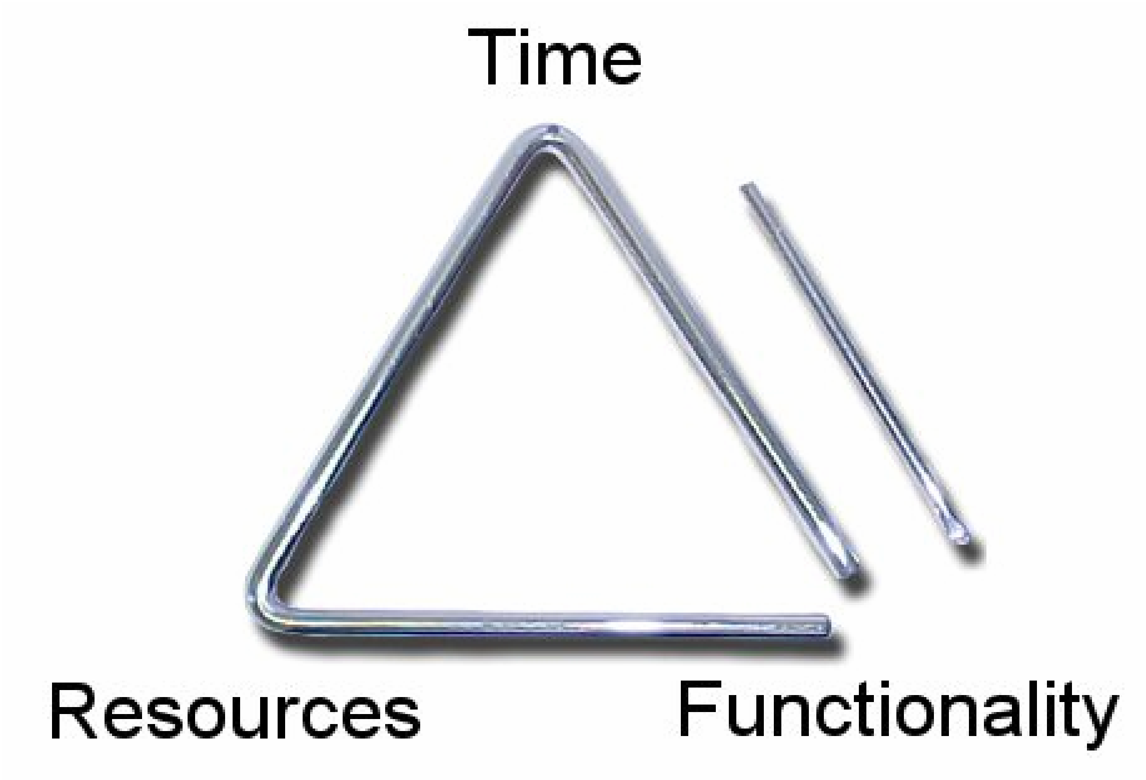
- ☑ Requirements in SCRUM
- ☑ Levels of Planning
- ☐ **Project Steering**
- ☐ Estimation Techniques

Steering your Projects

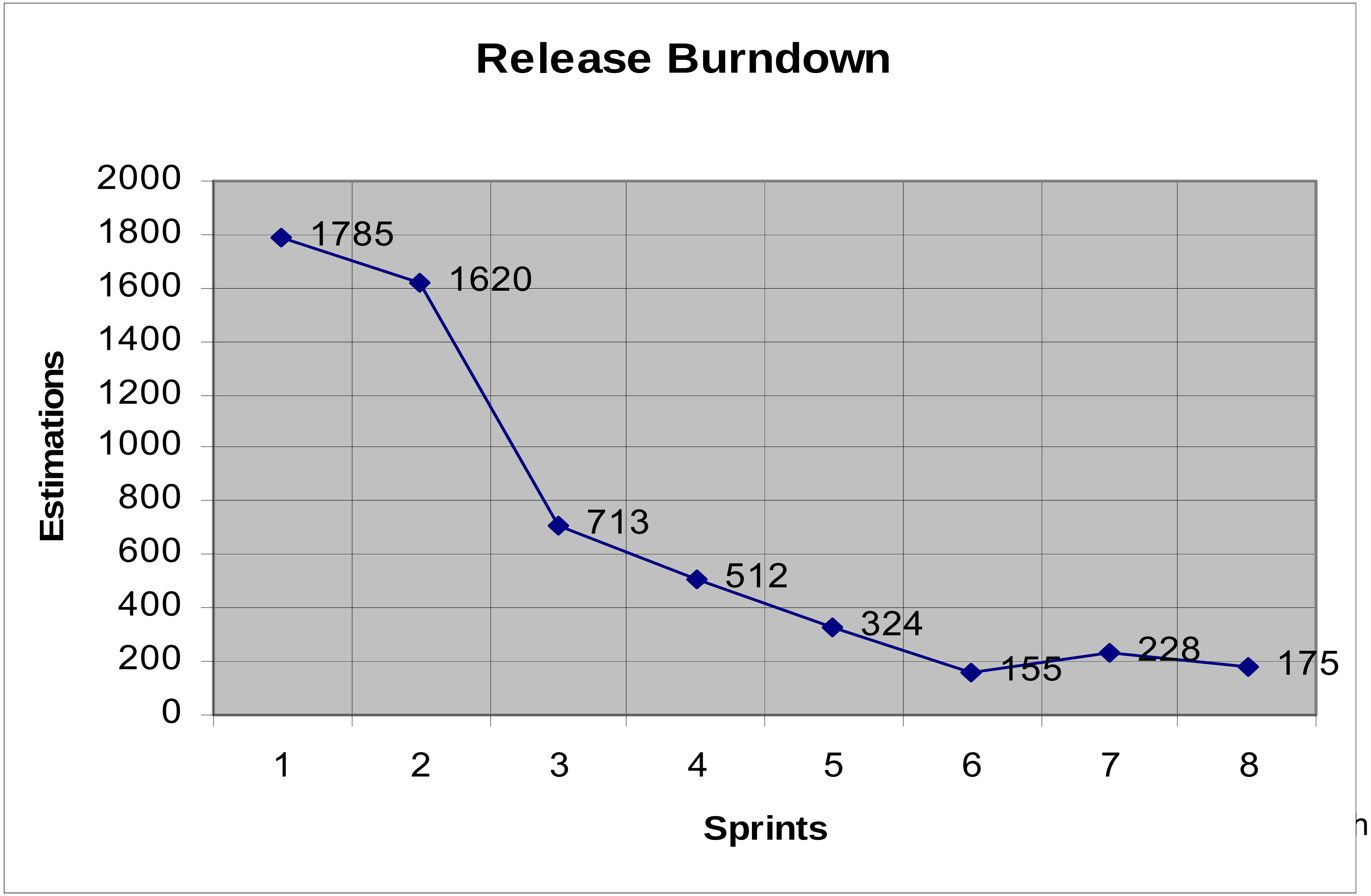
“Ready? Fire! Aim… Aim… Aim… Aim…”

© Kent Beck, Martin Fowler
“Planning eXtreme Programming”

There are four main parameters to planning



Visualizing the time-scope relationship



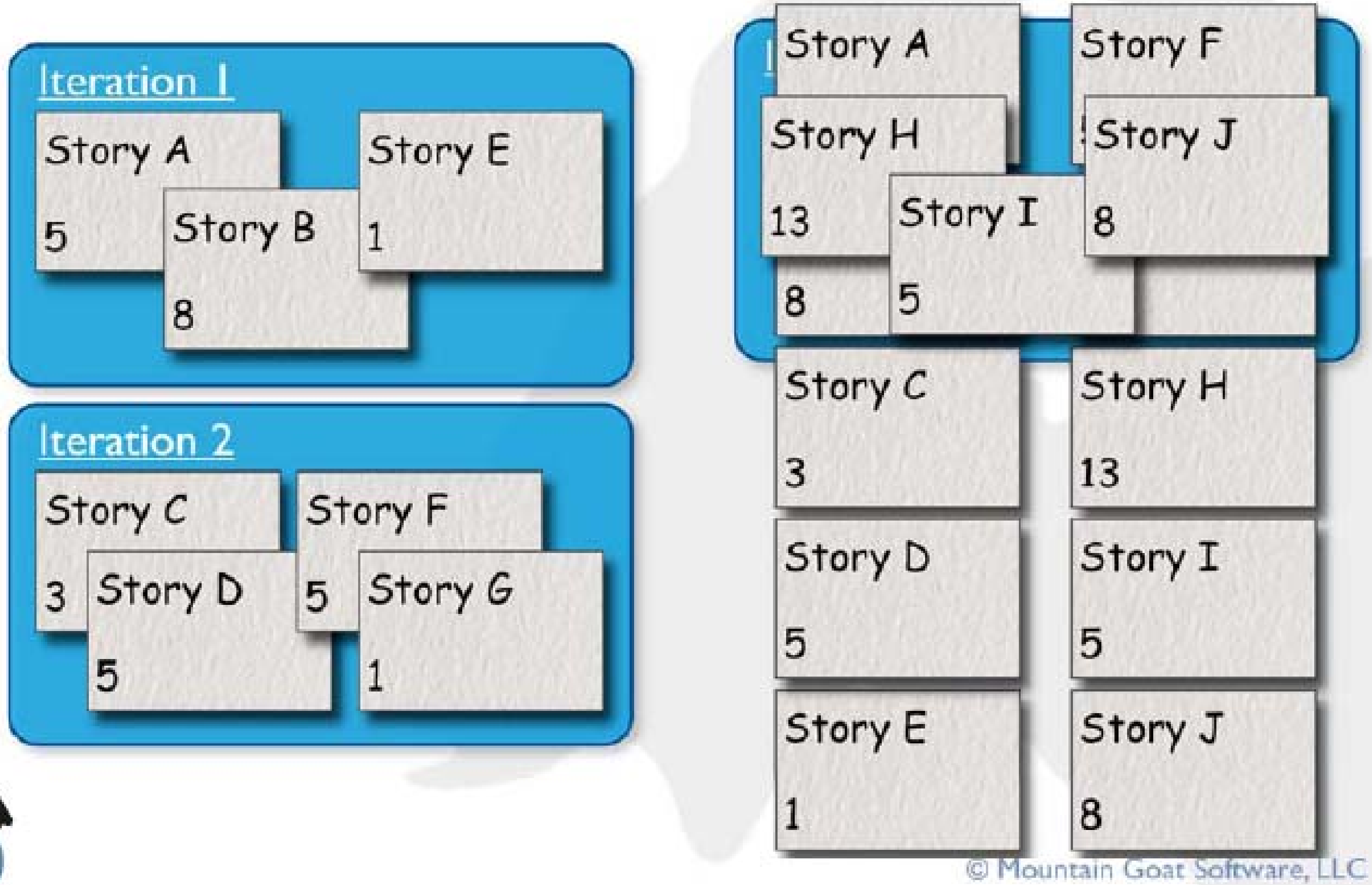
The Performance metric

Velocity – speed at which the team converts pieces of requirements into a <u>working</u> product during a single sprint.

Light and powerful metric.

From Mike Cohn's "Agile Estimating and Planning"

An example with velocity=14



Agile Planning

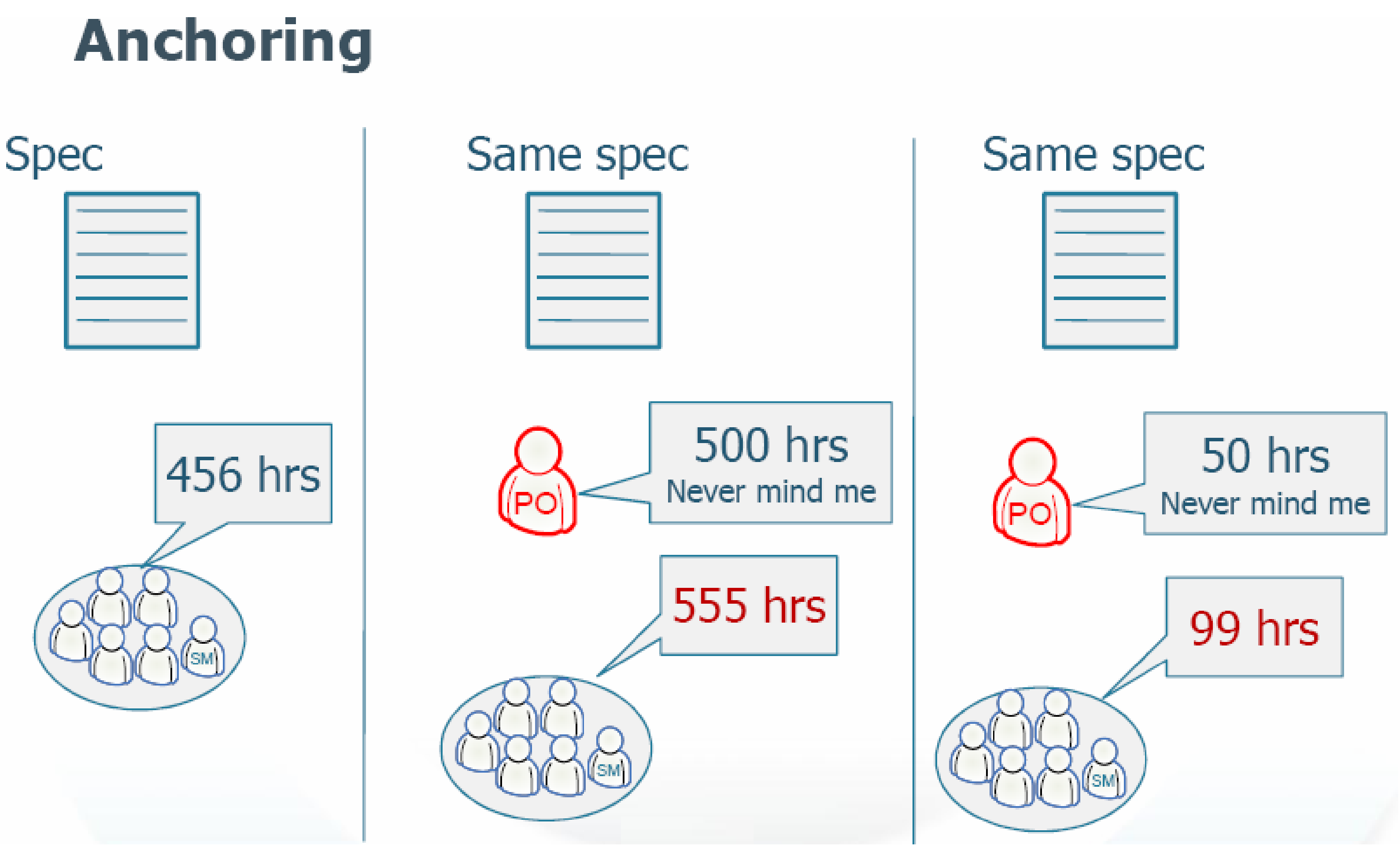
☑ Requirements in SCRUM

☑ Levels of Planning

☑ Project Steering

☐ **Estimation Techniques**

A typical issue with estimations



Planning Poker. The Steps

- Each estimator is given a deck of cards (1,2,3,5,8…)
- A Product Owner reads a story
- Estimators are for clarifications until everyone is ready to estimate
- Each estimator selects a card without showing it to the others
- By command cards are got turned over
- People with different estimates have discussions (limit with a timer)
- The round repeats until estimates converge

SCRUM and you

- In your current (next) project which SCRUM practices you think are the MUSTs?
- Which you find hard to do?
- Which you find impossible?
- Which you find unnecessary?
- Which can you start doing your next working day?

Questions? Concerns? Any feedback?

Thank you!

My contacts

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